# Midterm Exam I
## Project Progress Assessment

**Mode:** Group project
**Team size:** Final Project students
**Assessment type:** Functional MVP and technical presentation

### Objective

The objective of this midterm exam is to evaluate each team's ability to apply microservices concepts through the implementation of a small but functional solution based on the project selected by the group.

Each team must build or adapt a **frontend application** that consumes a provided **authentication API** for user login and integrates **at least one additional microservice** related to the domain of their chosen project. The goal is to demonstrate service separation, communication between components, and a basic but functional microservices-based MVP.

### General Instructions

Each project team must develop a functional microservices-based MVP with the following minimum scope:

1. A **frontend application** with a working **login screen**.
2. Integration with the provided **authentication API** so that users can log in through the frontend.
3. Implementation of **at least one additional microservice** based on the team's selected project idea.
4. The frontend must consume both:

    - the **authentication service**
    - the **additional project-related microservice**
5. The solution must demonstrate a clear microservices-oriented structure, with separated responsibilities between services and client application.

The purpose of the midterm is not to deliver the full final project, but to demonstrate a **functional initial MVP** that already shows the architectural direction of the team's solution.

**Minimum Technical Requirement**

At minimum, the team must demonstrate:

- a working **frontend login page**
- connection from the frontend to the **authentication API**
- successful authentication flow from the client side
- one additional **domain microservice** related to the project
- at least one functional flow in that second service that can be executed from the frontend after login
- source code stored in a **Git repository**
- evidence of work from all team members

**Examples of the second microservice**

The additional microservice depends on the team's project. Examples may include:

- student service
- appointment service
- reservation service
- inventory service
- incidents/reporting service
- academic request service
- notifications service
- products/orders service

The chosen service must be coherent with the problem the team selected for the course project.

**Expected MVP**

By the delivery date, each team must have a **functional MVP** in the repository.

At minimum, the MVP must allow:

- a user to access the frontend
- a user to perform login through the authentication API
- the frontend to call at least one additional microservice
- demonstration of one business functionality after login

The MVP does not need to be complete, but it must be functional, testable, and clearly represent the beginning of the final solution.

**Team Work Requirement**

All team members must participate in the development of the solution.

The contribution of each member will be reviewed through:

- commits in Git
- branch usage
- visible code contribution
- participation in implementation and presentation

Each student must be able to explain the part of the solution in which they participated.

**What Must Be Presented**

On the delivery date, each team must give a **PowerPoint presentation** and demonstrate the solution running.

The presentation must include:

- the problem addressed by the project

- the selected project context
- the proposed microservices-based solution
- the authentication flow
- the role of the frontend
- the purpose of the second microservice
- the current MVP scope
- the work distribution across the team

During the presentation, the team must also show:

- the **frontend login screen**
- the login working with the authentication API
- the second microservice working from the frontend
- the Git repository
- commit history and branch structure
- evidence of participation from all members
- the functional MVP running locally or deployed

**What Will Be Reviewed**

The instructor will verify:

- existence of a functional frontend
- correct use of the authentication API
- integration between frontend and services
- existence of at least one additional microservice
- coherence between the second microservice and the selected project
- functional MVP in the repository
- code organization and separation of responsibilities
- contribution from all team members
- quality and clarity of the final presentation
- technical understanding shown during the defense

**Deliverables / Evidence**

Each team must provide:

1. **A PowerPoint presentation**
2. **Link to the Git repository**
    - either one repository with all components
    - or separate repositories if the team decided to organize frontend and services independently
3. **A functional MVP available in the repository**

4. **A running demonstration** of the solution
5. **Visible Git history** showing team contribution
6. **A short README** explaining how to run the frontend and services locally
7. **A simple application architecture diagram** showing:
    - frontend
    - authentication API
    - second microservice
    - communication between them

### Suggested Architecture for the Midterm

The expected architecture can be simple, but it should clearly reflect microservices thinking.

At minimum, the solution should include:

- **Frontend**
- **Authentication API**
- **Project Domain Microservice**
- optional local storage or in-memory persistence
- communication through HTTP/REST

The goal is not to build a complex enterprise platform, but to show the initial implementation of a microservices-based solution.

### Evaluation Criteria

| Criterion | Weight |
|---|---|
| Project setup and repository structure | 10% |
| Frontend login integration with the authentication API | 25% |
| Implementation of the second project-related microservice | 25% |
| Integration between frontend and services / microservices design clarity | 15% |
| Contribution and participation of all team members in Git | 10% |
| Final presentation and functional demonstration | 15% |

**Important Evaluation Note**

This midterm is based not only on whether the application runs, but also on whether the team can demonstrate:

- clear service separation
- coherent architectural decisions
- real frontend-to-service interaction
- participation from all members
- understanding of the implemented solution

A team may lose points if:

- only the login works and no second service is implemented
- the second service is not related to the selected project
- the frontend does not actually consume the services
- only one team member appears to have done the work
- the team cannot explain the implemented architecture

**Final Note for Students**

This midterm is designed as the **first functional milestone** of your final course project. The purpose is to move from idea and design into an executable MVP that already reflects a microservices-based solution with authentication, frontend integration, and at least one domain service.